

Lymphatic & Immune Systems

A drainage network that doubles as a security network, and a defense that learns.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Say what lymph vessels return, and what happens when they are blocked
- Tell innate defense from adaptive defense on speed, aim, and memory
- Explain each sign of inflammation by what the vessels are doing
- Describe what B cells, helper T cells, and killer T cells do
- Explain why a second exposure is faster and bigger, and how vaccines use that

1 The drainage problem nobody mentions

Capillaries leak on purpose, and a little more fluid leaves than comes back.

That extra is a few liters a day. It would swell the tissues within hours if it stayed.

Lymph vessels collect it, filter it through lymph nodes, and empty it back into the big veins near the heart. There is no pump. Lymph moves by muscle squeezing and valves, just like blood in the veins.

Block the drainage and the fluid has nowhere to go. That is why removing or irradiating lymph nodes during cancer treatment can leave an arm or leg permanently swollen. The swelling is mechanical, not an infection.

Nodes are checkpoints, not just filters. Fluid from the tissues passes through crowds of immune cells before it rejoins the blood, so anything picked up in the tissue gets inspected on the way. That is why nodes near an infection swell and get sore, and why nodes are the first place doctors look for spreading cancer.

2 Two defense systems, working together

	INNATE	ADAPTIVE
Speed	Immediate	Days, the first time
Aim	Same response to anything	Aimed at one specific target
Memory	None	Yes, and that is the point
Players	Skin, mucus, eating cells, inflammation, fever	B cells, antibodies, helper and killer T cells

Innate defense buys time. Adaptive defense wins the fight and remembers how. Neither works well alone, and a vaccine is a rehearsal of the handover between them.

3 Inflammation, explained by the vessels

The four classic signs are not a list to memorize. They are one change in the vessels, seen four ways.

- 1 **Vessels widen.** More blood arrives, so the area is **red** and **warm**.
- 2 **Vessels get leakier.** Fluid and protein enter the tissue, so it **swells**.
- 3 **Swelling and chemicals irritate nerve endings.** So it **hurts**.
- 4 **Eating cells squeeze out** through those leaky walls and start clearing up debris and germs.

Every part of that is useful. More blood brings more cells and more clotting protein. The leak lets them out where they are needed. The pain forces rest.

Inflammation is only a problem when there is too much of it, or when it is aimed at the wrong thing.

Fever belongs here too, and it is a Module 1 idea. Chemicals from immune cells raise the set point in the brain. The body is not overheating. It is defending a new, higher

target, which is why the patient shivers and feels cold while the temperature climbs.

4 The adaptive cast

- **Helper T cells** are the coordinators. They spot the threat and switch on nearly everything else. Lose them, as in untreated HIV, and the whole adaptive system fails even though its other parts are fine.
- **Killer T cells** destroy the body's own cells once those cells are infected or cancerous.
- **B cells** are the factories. Once switched on, they pour out **antibodies** that travel in blood and lymph.
- **Memory cells** are why any of this matters a second time. Both B and T lines leave memory behind.

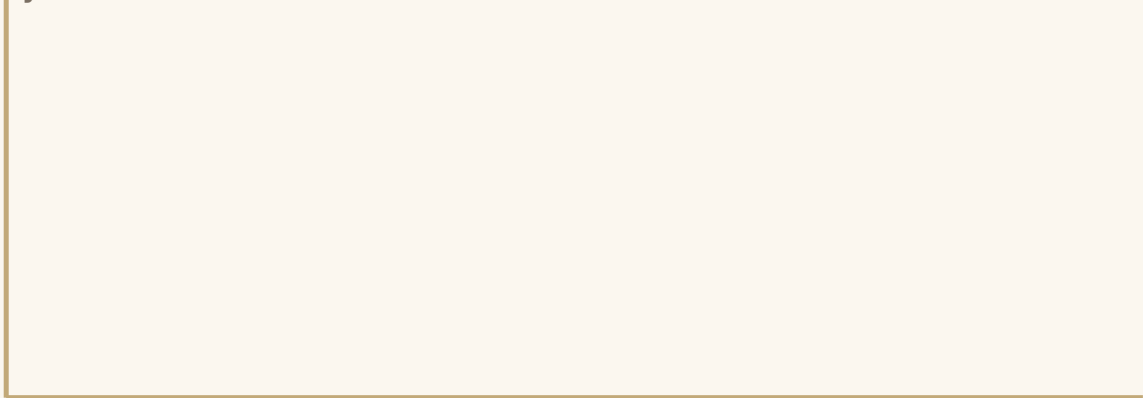
Antibodies do not kill anything themselves. They tag, clump, and block. Marking a target for destruction is the whole job.

5 Why the second exposure is different

Press the buttons and compare the two responses.

PRIMARY AND SECONDARY ANTIBODY RESPONSE

Antibody



Time

Press the button to meet this antigen for the first time, then press it again to meet it a second time and compare.

The first response is slow and small. The few cells that happen to recognize this target must be found, switched on, and multiplied.

The second is fast and huge, because memory cells are already there in numbers and skip the whole start-up.

That gap is what a vaccine buys you. It gives the body a harmless version of the target, so the slow, risky first response happens on a day when it does not matter. The memory is then in place for the day it does.

WHY THE NURSE CARES

ACTIVE

immunity means the patient's own system responded, either to an infection or to a vaccine. It is slow to arrive but it lasts.

PASSIVE

immunity means ready-made antibodies were given, as in an antibody treatment or those a newborn gets from its mother. It works immediately but fades, because no memory is made.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. Lymph nodes are removed from a patient's armpit during surgery, and the arm later swells. Why?

- a) The arm is infected
- b) Extra tissue fluid can no longer drain back to the blood
- c) Blood flow to the arm stopped
- d) The arm lost its vein valves

2. Which belongs to adaptive defense but not innate defense?

- a) Eating bacteria
- b) Memory of one specific target
- c) Skin as a barrier
- d) An immediate response

3. An inflamed area is red, warm, and swollen. What causes that directly?

- a) Vessels widen and get leakier
- b) White cells make heat as they divide
- c) Nerve endings release fluid
- d) Lymph vessels close

4. A patient shivers and feels cold while their temperature climbs to 39°C. Why?

- a) The thermostat broke and cannot read temperature
- b) The set point was raised, so the body is warming up to the new target
- c) Shivering has nothing to do with temperature control
- d) Vessels widened to lose heat

5. How does a vaccine protect against a disease the patient has never had?

- a) It gives ready-made antibodies that last for life
- b) It triggers a first response, so memory cells are ready before the real exposure
- c) It kills the germ on contact
- d) It permanently boosts innate defense

Answer key

1. **Extra tissue fluid can no longer drain back to the blood**

Correct. Capillaries leak more fluid than they take back, and lymph vessels are the way back.

2. **Memory of one specific target**

Correct. Specific memory is what makes the second exposure different.

3. **Vessels widen and get leakier**

Correct. More blood arriving gives red and warm. Leakier walls give swelling.

4. **The set point was raised, so the body is warming up to the new target**

Correct. Below the new set point, the body acts as though it is cold, because it now is.

5. **It triggers a first response, so memory cells are ready before the real exposure**

Correct. The slow, risky first response happens on a harmless day.

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Joyce University of Nursing and Health Sciences

Course resources by **Dr. Vidya Garg**